OpenDRIVE® 1.8.1, Static Road Network Description, 2024-11-21

11.2 Lane groups

For easier navigation through an ASAM OpenDRIVE road description, the lanes within a lane section are grouped into left, center, and right lanes.

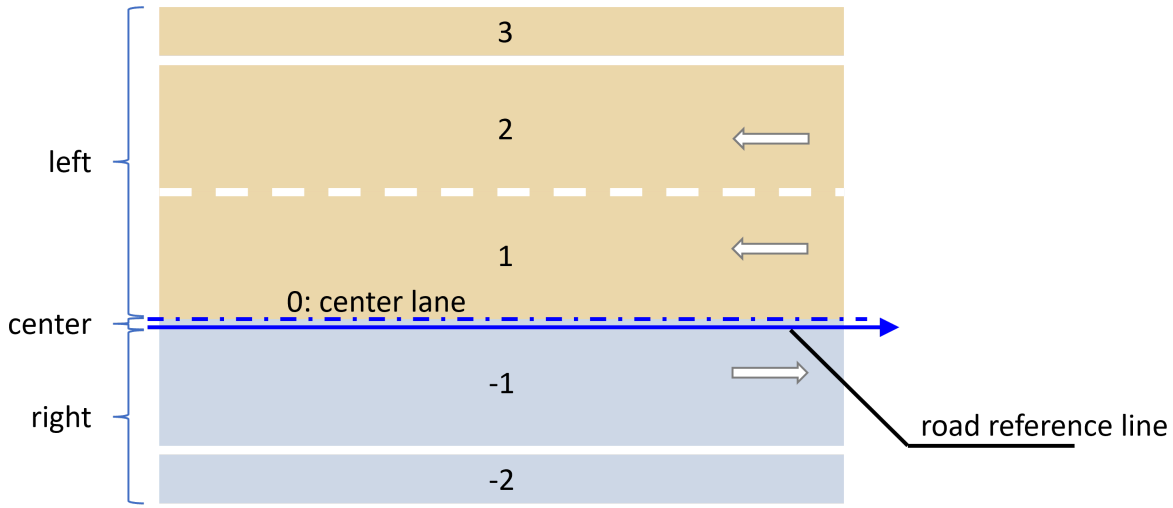

Figure 60. Lane grouping with left, center, right

Figure 60 shows the lane grouping. Within these groups, the lanes are described by `<lane>` elements. Because lane numbers descend in a negative t-direction and ascend in a positive t-direction, applications can derive the direction of a lane from the lane id given in the @id attribute of a `<lane>` element, unless the lane is bi-directional (specified by @direction=both).

Elements in UML model

`<left>` element

In ASAM OpenDRIVE, left lane groups are represented by the `<left>` element within the `<laneSection>` element.

```
UML class: t_road_lanes_laneSection_left
XML tag:   <left> (Multiplicity: 0..1)
```

Contains all lanes left to the center lane.

`<lane>` element

In ASAM OpenDRIVE, lanes in the left lane group are represented by `<lane>` elements within the `<left>` element.

```
UML class: t_road_lanes_laneSection_left_lane
XML tag:   <lane> (Multiplicity: 1..*)
```

Left lanes numbered with positive IDs in ascending order from center lane to left border.

Table 34. Attributes of the <lane> element

Name	Type	Use	Introduced	Description
`advisory`	e_laneAdvisory	optional	1.8.0	If true, lane can be used also by a neighboring lane. Advisory lane has priority, for example a bike lane, that can also be used by cars. If not specified, default value is none.
`direction`	e_lane_direction	optional	1.8.0	If not specified, direction is determined by the combination of `<left>` or `<right>` lane grouping and the values LHT or RHT of the @rule attribute of a road. The standard direction can be overwritten with this attribute.
`dynamicLaneDirection`	t_bool	optional	1.8.0	If true, lane direction can be changed dynamically by the scenario during the simulation. If not specified, default boolean value is false.
`dynamicLaneType`	t_bool	optional	1.8.0	If true, lane type can be changed dynamically by the scenario during the simulation. Typical example is a stop lane that can be changed by VMS boards to a driving lane. If not specified, default boolean value is false.
`id`	positiveInteger	required		ID of the lane

Name	Type	Use	Introduced	Description
`level`	t_bool	optional		"true" = keep lane on level, that is, do not apply superelevation; "false" = apply superelevation to this lane (default, also used if attribute level is missing)
`roadWorks`	t_bool	optional	1.8.0	If true, lane is under construction.
`type`	e_laneType	required		Type of the lane

`<center>` element

In ASAM OpenDRIVE, center lane groups are represented by the `<center>` element within the `<laneSection>` element.

```
UML class: t_road_lanes_laneSection_center
XML tag:   <center> (Multiplicity: 1)
```

Contains the center lane, which must be defined for all roads.

`<lane>` element

In ASAM OpenDRIVE, lanes in the center lane group are represented by `<lane>` elements within the `<center>` element.

```
UML class: t_road_lanes_laneSection_center_lane
XML tag:   <lane> (Multiplicity: 1)
```

Center lane element with ID zero. Has no width attribute. Mainly used for road marks.

Table 35. Attributes of the <lane> element

Name	Type	Use	Deprecated	Description
`id`	integer	required		ID of the lane

Name	Type	Use	Deprecated	Description
level	t_bool	optional	1.8.0	
type	e_laneType	optional	1.8.0	

<right> element

In ASAM OpenDRIVE, right lane groups are represented by the <right> element within the <laneSection> element.

```
UML class: t_road_lanes_laneSection_right
XML tag:   <right> (Multiplicity: 0..1)
```

Contains all lanes right to the center lane.

<lane> element

In ASAM OpenDRIVE, lanes in the right lane group are represented by <lane> elements within the <right> element.

```
UML class: t_road_lanes_laneSection_right_lane
XML tag:   <lane> (Multiplicity: 1..*)
```

Right lanes numbered with negative IDs in descending order from center lane to right border.

Table 36. Attributes of the <lane> element

Name	Type	Use	Introduced	Description
advisory	e_laneAdvisory	optional	1.8.0	If true, lane can be used also by a neighboring lane. Advisory lane has priority, for example a bike lane, that can also be used by cars. If not specified, default value is none.

Name	Type	Use	Introduced	Description
`direction`	e_lane_direction	optional	1.8.0	If not specified, direction is determined by the combination of `<left>` or `<right>` lane grouping and the values LHT or RHT of the @rule attribute of a road. The standard direction can be overwritten with this attribute.
`dynamicLaneDirection`	t_bool	optional	1.8.0	If true, lane direction can be changed dynamically by the scenario during the simulation. If not specified, default boolean value is false.
`dynamicLaneType`	t_bool	optional	1.8.0	If true, lane type can be changed dynamically by the scenario during the simulation. Typical example is a stop lane that can be changed by VMS boards to a driving lane. If not specified, default boolean value is false.
`id`	negativeInteger	required		ID of the lane
`level`	t_bool	optional		"true" = keep lane on level, that is, do not apply superelevation; "false" = apply superelevation to this lane (default, also used if attribute level is missing)
`roadWorks`	t_bool	optional	1.8.0	If true, lane is under construction.
`type`	e_laneType	required		Type of the lane

Rules

The following rules apply to lane grouping:

- asam.net:xodr:1.4.0:road.lane.lanes_numbered_correctly: Lanes with positive ID run on the left side of the center lane, while lanes with negative ID run on the right side of the center lane.
- In order to be drivable, each lane section should contain at least one <right> or <left> element.
- asam.net:xodr:1.4.0:road.lane.center_elem_definition: One `<center>` element shall be defined for each s-coordinate.
- asam.net:xodr:1.4.0:road.lane.lane_listing: For better orientation, lanes should be listed from left to right, that is with descending ID.

- @direction="reverse" shall not be used to change from right-hand traffic to left-hand traffic and vice versa.

Related topics

- Section 11.1, "Introduction to lanes"
- Section 11.3, "Lane sections"

11.2.1 Driving direction

In ASAM OpenDRIVE, the driving direction is specified by a combination of different elements and attributes. For a road with the @rule="RHT" attribute, the default driving direction would be in positive direction of the road reference line for all `<right>` element lanes with negative @id attribute and against the road reference line for lanes in the `<left>` element with positive @id attribute. If the road has the @rule="LHT" attribute, the default driving direction would be in positive direction of the road reference line for all `<left>` element lanes with positive @id attribute and against the road reference line for all `<right>` element lanes with negative @id attribute. This can be influenced with the @direction attribute individually for each lane. If the @direction attribute is not specified or has a value of @direction="standard", the default driving direction is not changed. The @direction="reversed" attribute reverses the default driving direction. The @direction="both" attribute replaces the deprecated lane @type="bidirectional" and allows both driving directions.

In addition, this can be changed during the simulation if @dynamicLaneDirection="true" attribute is set, for example by a VMS board.

Related topics

- Section 10.1, "Introduction to roads"
- Section 14.7, "Signal boards"